

1. Administrative & Study Identification

Full Study Title: A Phase 3, Multicenter, Open-label, Randomized Study to Compare the Efficacy and Safety of MK-2870 Versus Treatment of Physician's Choice in 3L+ Advanced/Metastatic Gastroesophageal Adenocarcinoma (Gastric Adenocarcinoma, Gastroesophageal Junction Adenocarcinoma, and Esophageal Adenocarcinoma) **Short Title/Acronym:** MK-2870-015 **Protocol Number:** MK-2870-015 **NCT Number:** NCT06356311 **Sponsor Name:** Merck **Investigational Product:** Sacituzumab Tirumotecan (MK-2870) **Phase of Development:** Phase III **Trial Status:** Recruiting **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate and compare the efficacy of sacituzumab tirumotecan (MK-2870) versus the Treatment of Physician's Choice (TPC) in demonstrating superiority with respect to Overall Survival (OS) in participants with advanced/metastatic gastroesophageal adenocarcinoma. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Objective Response Rate (ORR), and Duration of Response (DOR).

2.2 Background & Rationale To address the critical unmet medical need for effective later-line therapies in advanced/metastatic gastroesophageal cancers. Patients who have progressed on at least 2 prior chemotherapy and/or immunotherapy regimens (3L+ setting) face a poor prognosis. This study evaluates MK-2870, a trophoblast cell-surface antigen 2 (TROP2) targeted antibody-drug conjugate (ADC), to determine if it provides a superior survival benefit over standard-of-care physician's choice therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Treatment of Physician's Choice / TPC) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 450 evaluable patients **Number of Sites:** Multicenter (Global) **Current Status:** Recruiting (Currently recruiting participants) **Estimated Primary Completion:** [Pending / Event-Driven]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically-or cytologically-confirmed diagnosis of advanced, unresectable or metastatic gastric adenocarcinoma, gastroesophageal junction adenocarcinoma, or esophageal adenocarcinoma. 2. Has received, and progressed on, at least 2 prior chemotherapy and/or immunotherapy regimens (3L+). Participants are eligible regardless of HER2 status, but HER2+ participants must have previously received trastuzumab. 3. Has measurable disease per RECIST 1.1 and an ECOG performance status of 0 or 1. 4. Provided tumor tissue sample for determination of TROP2 status by the central laboratory before randomization. 5. Has the ability to swallow oral medication for those who may receive trifluridine-tipiracil. 6. HIV, HBV, or HCV infected participants are eligible if well-controlled on appropriate therapy (ART or antivirals) and have undetectable viral loads.

4.2 Exclusion Criteria 1. Prior treatment with a TROP2 targeted ADC, a topoisomerase 1 inhibitor, or a topoisomerase 1 inhibitor-based chemotherapy. 2. Active central nervous system (CNS) metastases, carcinomatous meningitis, or history of noninfectious pneumonitis/interstitial lung disease (ILD) that required steroids. 3. Uncontrolled cardiovascular disease (e.g., NYHA Class III/IV, QTcF > 480 ms), active inflammatory bowel disease, or severe dry eye syndrome/corneal disease. 4. Grade $\geq$ 2 peripheral neuropathy. 5. Accumulation of pleural, ascitic, or pericardial fluid requiring drainage or diuretic drugs within 2 weeks before the first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-2870 at specified weight-based dosing intervals OR Treatment of Physician's Choice (e.g., oral trifluridine-tipiracil). (*Note on TPC drug details: Trifluridine [ATC Code: S01AD02; Trade Names: Viroptic, Lonsurf]. EMA Product Information available at: https://www.ema.europa.eu/en/documents/product-information/lonsurf-epar-product-information_en.pdf*). **Route of Administration:** IV (for MK-2870 and IV TPC regimens) and/or Oral (for oral TPC regimens). **Duration of Treatment:** Continued until documented disease progression per RECIST 1.1, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care; hormone replacement therapy for adequately treated endocrine-related AEs. **Prohibited:** Strong and/or moderate inducers/inhibitors of CYP3A4 (requires 2-week washout); live or live-attenuated vaccines within 30 days prior to first dose; prior systemic anticancer therapy or radiotherapy within 2 weeks of the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor tissue submission for TROP2 status, Baseline Imaging (RECIST 1.1)
Baseline	Day 0	Randomization, First Dose
Interim	Every Cycle (e.g., Q2W/Q3W)	Safety Labs, Treatment Administration, AE Monitoring
End of Study/Follow-Up	End of Treatment	Tumor imaging, Survival Follow-up (OS monitoring), End of Treatment Safety Labs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS): defined as the time from randomization to death due to any cause.

Measurement Method: Standard mortality tracking and survival follow-up assessments.

7.2 Secondary & Exploratory Endpoints 1. Progression-free survival (PFS) 2. Objective Response Rate (ORR) 3. Duration of Response (DOR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine laboratory evaluations, physical exams, and patient reporting, with special attention to monitoring for ILD/pneumonitis, peripheral neuropathy, and ocular toxicities. **Serious Adverse Events (SAEs):** Must be reported to the sponsor and regulatory authorities within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) will evaluate ongoing safety and efficacy data across treatment arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy endpoints (OS, PFS, ORR, DOR); Safety Set for all participants receiving ≥ 1 dose of study drug. **Sample Size Justification:** $N = 450$ is sized to provide adequate statistical power ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to detect a statistically significant superiority in Overall Survival for MK-2870 compared to TPC. **Handling of Missing Data:** Time-to-event endpoints (OS, PFS, DOR) will utilize standard Kaplan-Meier estimates and appropriate censoring rules for patients lost to follow-up or without an event.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard risk-based monitoring, including regular onsite and remote data verification. **Audit Trail:** Robust Electronic Data Capture (EDC) system with automated audit trails, eCRF locking, and rigorous data clarification workflows.

11. Study Identifiers & Governance

Primary/Posting ID: 511990488076540549 **Registry Number:** NCT06356311 **Sponsor/Lead Organization (Company):** Merck **Study Title:** A Study to Evaluate Sacituzumab Tirumotecan (MK-2870) in Advanced/Metastatic Gastroesophageal Adenocarcinoma (MK-2870-015) **Official Regulatory Title:** A Phase 3, Multicenter, Open-label, Randomized Study to Compare the Efficacy and Safety of MK-2870 Versus Treatment of Physician's Choice in 3L+ Advanced/Metastatic Gastroesophageal Adenocarcinoma (Gastric Adenocarcinoma, Gastroesophageal Junction Adenocarcinoma, and Esophageal Adenocarcinoma) **Source Level:** 5

12. Clinical Framework (Oncology Specific)

Disease Areas: Gastroesophageal Cancer **Primary Condition:** Advanced/Metastatic Gastroesophageal Adenocarcinoma **Preferred UMLS Name:** Gastroesophageal Adenocarcinoma **Preferred Disease Name:** Malignant neoplasm of cardioesophageal junction of stomach **Diagnosis Threshold:** Histologically/cytologically-confirmed progressing disease (3L+) **Medical Methodology:** Targeted Antibody-Drug Conjugate vs. Standard Chemotherapy **Optimization Target:** Improved Overall Survival (OS) and delayed disease progression

13. Study Procedures & Methodology

Management Pathway: 3rd-Line+ Systemic Anti-Cancer Therapy **Education Protocol (HCP):** Protocol training, RECIST 1.1 response evaluation standardization, and ILD/Pneumonitis monitoring guidelines **Education Protocol (Patient):** Informed consent, AE reporting self-awareness, and oral therapy adherence training (if randomized to oral TPC) **Quality Audit Mechanism:** Independent Central Radiologic Review (ICR) for objective tumor assessments

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until death, withdrawal of consent, or study termination **Onsite Visit Cadence:** Pre-dose on Day 1 of every treatment cycle **Remote Monitoring Frequency:** Inter-cycle

telemedicine for symptom management or Survival Follow-Up phase (e.g., Q12W) **Checklist Review Interval:** End of every treatment cycle for dose-limiting toxicities

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					